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Key

## SKILLS QUIZ

1. a) Carry out the following matrix-vector multiplication.

$$\begin{pmatrix} -1 & 2 & 4 & 3 \\ 1 & 0 & 8 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -1+6+0+9 \\ 1+0+0-3 \end{pmatrix} = \begin{pmatrix} 14 \\ -2 \end{pmatrix}$$

b) Give an example of a matrix that represents a linear transformation from  $\mathbb{R}^4$  to  $\mathbb{R}^3$  (with the understanding that the standard bases are being used); or explain why this is not possible.

Any  $3 \times 4$  matrix with real entries is  
a correct answer. For instance,

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 9 & 12 & 15 & 18 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

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## SKILLS QUIZ

2. a) For each of the following matrix operations, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 0 & 1 & 4 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 2 & 4 \end{pmatrix}$$

$A+B$ ,  $CD$ ,  $CA$ ,  $DA$

$$A+B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$CD = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 0 & 1 & 4 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 0 & 4 \end{pmatrix}$$

$CA$  is not defined.

$$DA = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ -2 & 4 \end{pmatrix}$$

b) Give an example of two matrices  $A$  and  $B$  so that their product  $AB$  is a zero matrix, but so that their sum  $A+B$  is not defined; or explain why this is not possible.

If  $A+B$  is not defined, then  $A$  and  $B$  must have different dimensions. If  $AB$  is defined, then  $A$  is  $m \times n$  and  $B$  is  $n \times k$  for  $m, n, k$  positive whole numbers. So as long as  $m, n, k$  are not all equal,  $AB$  is defined but  $A+B$  is not.

So for instance, examples include:

$$\begin{matrix} A & B \\ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \text{or} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 9 & 2 & 6 & 5 \\ 0 & 1 & 0 & 8 \end{pmatrix} \end{matrix}$$

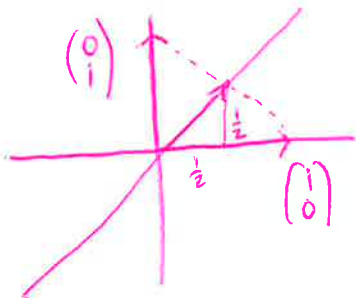
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## SKILLS QUIZ

3. a) Find the  $2 \times 2$  matrix for the following linear transformation using the standard basis for  $\mathbb{R}^2$ .

project to the line  $y = x$  and then rotate by  $90^\circ$  counterclockwise



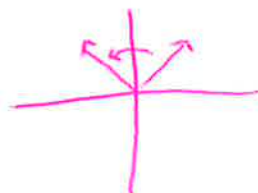
Project to  $y=x$ :

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

And under  $90^\circ$  CCW rotation,

$$\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} \mapsto \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix}$$



So the composition yields the matrix

$$\begin{pmatrix} -1/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

- b) Find the  $3 \times 3$  matrix for the following linear transformation using the standard basis for  $\mathbb{R}^3$ .

rotate  $180^\circ$  about the  $x$ -axis

fixed by the rotation

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

notice: this submatrix is the  $180^\circ$  rotation matrix for the  $yz$ -plane.

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## SKILLS QUIZ

4. a) Use row reduction to put the following real matrix in reduced row echelon form.

$$\begin{pmatrix} 0 & 10 & 10 & 10 \\ 2 & 4 & 0 & 10 \\ 0 & 6 & 8 & 10 \end{pmatrix} \xrightarrow{R1 \leftrightarrow R2} \begin{pmatrix} 2 & 4 & 0 & 10 \\ 0 & 10 & 10 & 10 \\ 0 & 6 & 8 & 10 \end{pmatrix} \xrightarrow{\frac{1}{2}R1} \begin{pmatrix} 1 & 2 & 0 & 5 \\ 0 & 10 & 10 & 10 \\ 0 & 6 & 8 & 10 \end{pmatrix}$$

$$\xrightarrow{\frac{1}{10}R2} \begin{pmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 6 & 8 & 10 \end{pmatrix} \xrightarrow{R3 - 6R2} \begin{pmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{\frac{1}{2}R3} \begin{pmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow{R2 - R3} \begin{pmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R1 - 2R2} \begin{pmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

b) Represent the following linear system over  $\mathbb{R}$  using a matrix. Then find the general solution to the system using row reduction and write the general solution in vector form.

no need to augment,  
since homogeneous.

$$\begin{aligned} w + 2x + 4y - 2z &= 0 \\ w + x + 3y + 4z &= 0 \end{aligned}$$

(But it is also  
fine to augment;  
you just carry around  
a lot of zeros.)

$$\begin{pmatrix} 1 & 2 & 4 & -2 \\ 1 & 1 & 3 & 4 \end{pmatrix}$$

$$\xrightarrow{R2 - R1} \begin{pmatrix} 1 & 2 & 4 & -2 \\ 0 & -1 & -1 & 6 \end{pmatrix} \xrightarrow{-R2} \begin{pmatrix} 1 & 2 & 4 & -2 \\ 0 & 1 & 1 & -6 \end{pmatrix}$$

$$\xrightarrow{R1 - 2R2} \begin{pmatrix} 1 & 0 & 2 & 10 \\ 0 & 1 & 1 & -6 \end{pmatrix} \xrightarrow{\text{RREF}}$$

$$\begin{aligned} w + 2y + 10z &= 0 \\ x + y - 6z &= 0 \end{aligned} \quad \Rightarrow \quad \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -10 \\ 6 \\ 0 \\ 1 \end{pmatrix}$$

vector form. of solution set.

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## SKILLS QUIZ

5. a) Compute the inverses of the following matrices; or explain why this is not possible.

$$\begin{pmatrix} 10 & 6 \\ 2 & 1 \end{pmatrix} \quad \begin{pmatrix} 4 & 0 & 1 \\ 0 & 0 & -1/2 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 10 & 6 \\ 2 & 1 \end{pmatrix}^{-1} = \frac{-1}{-2} \begin{pmatrix} 1 & -6 \\ -2 & 10 \end{pmatrix} = \begin{pmatrix} -1/2 & 3 \\ +1 & -5 \end{pmatrix} = \begin{pmatrix} -1/2 & 3 \\ 1 & -5 \end{pmatrix}$$

since determinant of the original matrix is  $-2$ ,

$$\frac{1}{\det A} = -\frac{1}{2}.$$

b) Always, sometimes, or never: an invertible matrix and its inverse both have the same nullspace.  
(If always or never, prove it. If sometimes, give an example each way.)

$$\begin{array}{ccc|ccc} 4 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1/2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \xrightarrow{R2 \leftrightarrow R3} \begin{array}{ccc|ccc} 4 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1/2 & 0 & 1 & 0 \end{array} \xrightarrow{-2R3} \begin{array}{ccc|ccc} 4 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -2 & 0 \end{array}$$

$$\begin{array}{ccc|ccc} 4 & 0 & 1 & 1 & 2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -2 & 0 \end{array} \xrightarrow{R1 - R3} \begin{array}{ccc|ccc} 4 & 0 & 0 & 1 & 4 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -2 & 0 \end{array} \xrightarrow{\frac{1}{4}R1} \begin{array}{ccc|ccc} 1 & 0 & 0 & 1/4 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -2 & 0 \end{array}$$

$$\text{So } \begin{pmatrix} 4 & 0 & 1 \\ 0 & 0 & -1/2 \\ 0 & 1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 1/4 & 1/2 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & 0 \end{pmatrix}$$

5b) Always. If  $A$  is invertible, it has full rank and has only the trivial vector in its nullspace. (The trivial vector in the domain of  $A$ .) And  $A^{-1}$  is also invertible (its inverse is  $A$ ), so it also has full rank and again its nullspace equals the trivial vector.